Sizing a Rooftop Solar Array

Finding the area of an oblique triangle from survey bearings

Group activity — advanced version

$\frac{1}{2}ab\sin C$ applied to a real total-station survey

The brief

One number decides the whole design

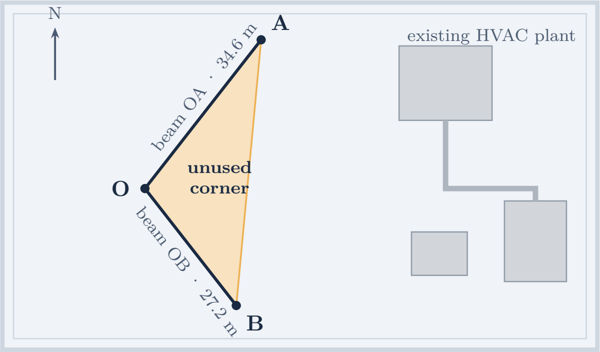
An engineering firm is scoping a small solar array in an unused triangular corner of a warehouse rooftop, boxed in by two existing structural beams. They need **the enclosed area** — that is what sets how many panels the site can hold.

The catch: the surveyor never recorded an angle. The field notes contain **two bearings**, which is how rooftop and land surveys are actually reported. So the job splits in two — recover the angle, then find the area.

1. The site and the field notes
2. Bearings $\rightarrow$ included angle
3. The triangle, reduced to SAS
4. Why $\frac{1}{2}ab\sin C$ works
5. The calculation
6. The answer, and what it is worth

The site

Plan view of the rooftop corner



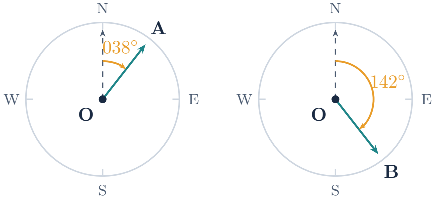
Plan view · warehouse rooftop bounded by two structural beams

What the surveyor actually recorded

Total station, set up over point O

Field notes — station O

Beam OA	34.6 m at bearing 038°
Beam OB	27.2 m at bearing 142°
Angle at O	*not recorded*



A bearing is always measured *clockwise from north* — same origin, same sense, every time.

A total station measures **distance and direction**, and direction is logged as a bearing — nobody stands in the corner with a protractor. **So the included angle has to be derived.**

From two bearings to one angle

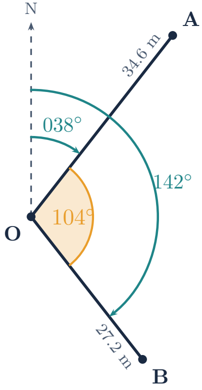
Both are measured from the same point, in the same sense

Two conditions make this a plain subtraction:

- both bearings share the **same reference point**, O;
- both are measured **clockwise from north**.

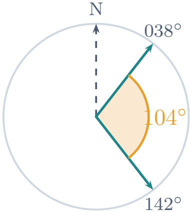
$$\angle AOB = 142^\circ - 38^\circ = \mathbf{104^\circ}$$

No trigonometry yet — this step is bookkeeping about how the directions were recorded.



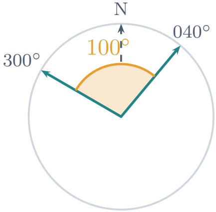
The rule in general

Subtraction is not always the whole story



$142 - 38 = 104 < 180$

take it directly



$300 - 40 = 260 > 180$

use $360 - 260 = 100°$

Subtract the bearings; if the result exceeds 180° you have measured the *reflex* side, so take 360° minus it instead. Here $104° < 180°$, so no adjustment is needed.

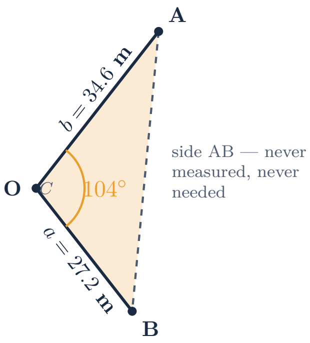
Now it is a standard SAS triangle

Two sides and the angle between them

$$
\begin{aligned}
a &= \mathrm{OB} = 27.2\ \mathrm{m} \\
b &= \mathrm{OA} = 34.6\ \mathrm{m} \\
C &= 104^\circ\ \text{(derived)}
\end{aligned}
$$

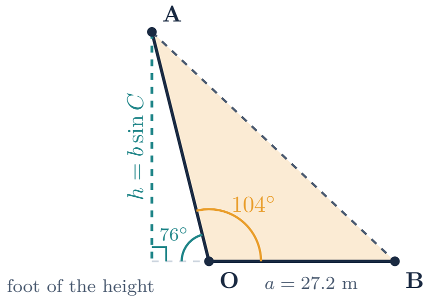
The angle sits **between** the two known sides — exactly the SAS pattern.

C is obtuse, so the triangle is genuinely oblique: no right angle to fall back on, and no perpendicular height in the field notes.



Why $\frac{1}{2}ab\sin C$ is the right tool

sin C recovers the height nobody measured



$$\text{Area} = \tfrac{1}{2} \times \text{base} \times \text{height}$$
$$= \tfrac{1}{2}\, a\,(b \sin C)$$

C is obtuse, so the foot lands outside the triangle — and $\sin 104° = \sin 76°$ still holds.

The familiar $\frac{1}{2} \times$ base $\times$ height never went away. **$\sin C$ simply supplies the height** from the side and angle we do have.

The calculation

Substitute once, then evaluate

$$\begin{aligned}
\text{Area} &= \tfrac{1}{2}\, ab \sin C \\
&= \tfrac{1}{2} \times 34.6 \times 27.2 \times \sin 104^\circ \\
&= 470.56 \times \sin 104^\circ \\
&= 470.56 \times 0.9703 \\
&\approx \mathbf{456.58}\ \mathrm{m}^{\mathbf{2}}
\end{aligned}$$

Watch for

Calculator in **degrees**, not radians.

$\sin 104^\circ = \sin 76^\circ = 0.9703$ — an obtuse angle gives a perfectly ordinary positive sine.

Round only at the **final step**.

Complete solution

Every quantity, in order

Quantity	Value
Side OA (b)	34.6 m
Side OB (a)	27.2 m
Bearing to A	038°
Bearing to B	142°
Included angle C	$142° - 38° = 104°$
$\sin 104°$	≈ 0.9703
$\frac{1}{2} \times$ OA $\times$ OB	470.56
Area	$\approx$ **456.58** m$^{\mathbf{2}}$

The triangular rooftop section available for solar panels is

$$456.58\ \mathbf{m}^2$$

from two beam lengths and two bearings — no angle ever measured

A note on rounding

Where the last decimal comes from

$\sin 104^\circ = 0.970296\ldots$, so

$$470.56 \times 0.970296 = 456.582\ldots$$

A rounded 0.9709 would give 456.87 m^2 — about 0.3 m^2 adrift, roughly **one panel's worth** of area.

Small, but the kind of error that compounds: carry the full value through and round once, at the end.

Sensitivity of the answer

$\sin C = 0.9703$	456.58 m^2
$\sin C = 0.9709$	456.87 m^2

difference: 0.29 m^2

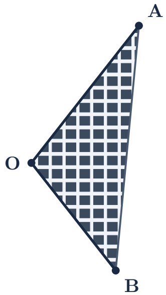
Optional: what could that area produce?

Flag this out loud as an estimate

Utility-scale arrays are often quoted at roughly 170 W of installed capacity per square metre:

$$456.58\ \mathrm{m}^2 \times 170\ \mathrm{W/m}^2 \approx 77\,619\ \mathrm{W} \approx \mathbf{77.6}\ \mathrm{kW}$$

Say this out loud: a rough, order-of-magnitude estimate — not an engineering calculation. It ignores tilt, row spacing, shading and inverter losses, all of which reduce the real figure.



schematic only — tilt, row spacing and shading are not modelled

What to take away

Two ideas, one method

1. The angle was earned, not given

Bearings share a reference direction and a rotational sense, so their difference *is* the angle between the lines. Real survey data arrives this way; converting it is part of the problem.

2. It is still just base $\times$ height

$\frac{1}{2}ab\sin C$ is not a new formula to memorise. $\sin C$ stands in for a perpendicular height that was never measured — which is why SAS is enough.

From **two distances and two directions** to **456.58 m^2** — and a panel count.

Questions?